

System Design Description

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INTRODUCTION

DOCUMENT PURPOSE

The document contains the design of the library management system and a description of its structure and operation. The document is intended for system developers and testers.

SCOPE OF OPERATION

The system is being developed with the aim of making library operations easier for both library staff and users, promoting effective information exchange and providing the opportunity to maintain and expand the resources offered by the library.

DEFINITIONS AND ABBREVIATIONS

Table 1

Term name	Term explanation
User	A person who wants to receive information using the program and provides data for operations.
Design document	A document that describes in detail the system's structure and operation so that, by following this information, the system can be implemented according to the specified requirements.
Decomposition	The division of a system into multiple components, for example modules, procedures, etc.
SQLite	A database management system that allows the development and maintenance of a database, as well as the creation of internal applications for working with the database.
Module	A component of a system that performs a specific function and operates as part of a larger system.
Process	Within a system, an identifiable sequence of specific data-processing activities with a defined initial state and possible end states.

RELATIONSHIP WITH OTHER DOCUMENTS

The document is based on the system requirements specification document and materials presented in the lessons on creating a design document.

DESCRIPTION OF DECOMPOSITION

MODULAR DECOMPOSITION

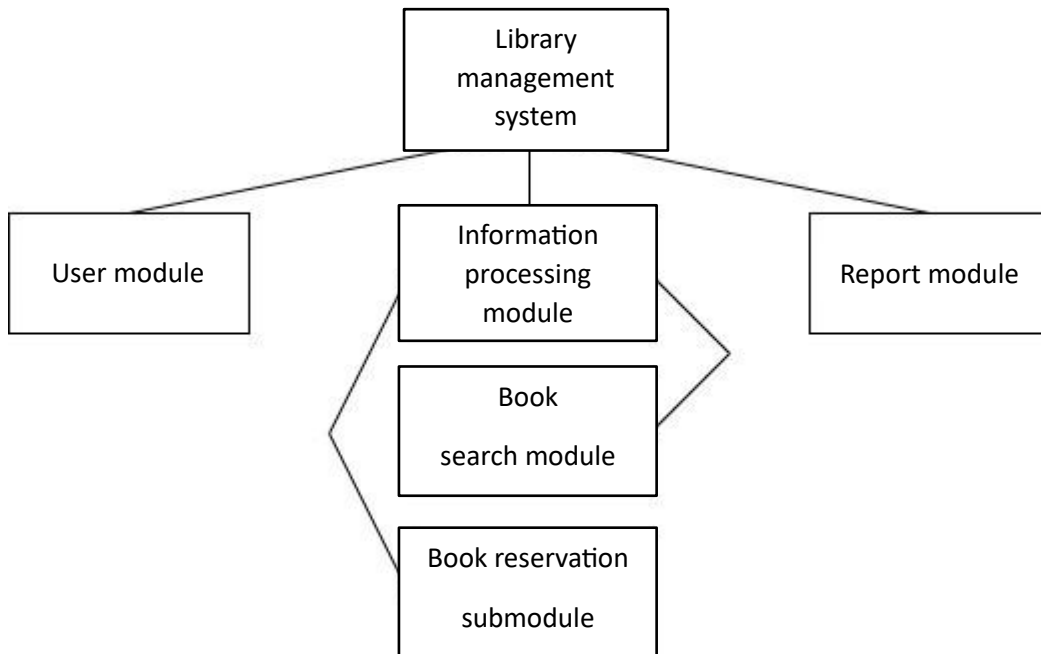


Image 1 – Modular Decomposition

MODULE DECOMPOSITION LIST

Table 2

Name	User Module
Type	User Module
Purpose	Data usage and service reception.
Function	To view information, perform searches, and make reservations.

Table 3

Name	Collection Processing Module
Type	Collection Processing Module
Purpose	Submodule selection.
Function	Selects the appropriate submodule depending on the required functionality.

Table 4

Name	Book Search Submodule
Type	Collection Processing Module

Purpose	Working with the book database.
Function	Processes the book title entered by the user and searches the database. Displays book information on the screen.

Table 5

Name	Book Addition and Removal Submodule
Type	Collection Processing Module
Purpose	Working with the library collection.
Function	Adds new information to the database and removes unnecessary or reserved books from the list of available books.

Table 6

Name	Book Reservation Submodule
Type	Collection Processing Module
Purpose	Managing reservations.
Function	Submits a reservation request for a book and checks its availability.

Table 7

Name	Reporting Module
Type	Reporting Module
Purpose	Generation of predefined reports.
Function	Uses predefined SQL queries to display book data on the screen or to show an error message if an issue occurs.

PROCESS DECOMPOSITION

Table 8

User Module	
Process Name	Process Description
Reservation Request	Submits a reservation for the selected book if it is available.
Collection Search	Enters the book title to obtain information about it in the database.

Table 9

Module	Process Name	Process Description
Collection Processing Module	Submodule Selection	Selects the appropriate submodule depending on the required functionality.

Book Search Submodule	Information Search in Database	Based on the titles entered by the user, the system checks for matching records using SQL queries. It then displays information about the book requested by the user.
Book Addition and Removal Submodule	Adding a Book to the Database	Library staff use SQL queries to add a record and additional information about the book to the database.
Book Addition and Removal Submodule	Removing a Book from the Database	Library staff use SQL queries to delete a record from the database.
Book Reservation Submodule	Assigning Reservation Status to a Book	Library staff use SQL queries to assign a reservation status to a book and make it unavailable to other users.

Table 10

Module	Process Name	Process Description
Reporting Module	Generation of Predefined Reports	Creates SQL queries for the required collection-related operations and displays the information on the screen.

DATA DECOMPOSITION

Table 11

Database Table Name	Description
books	The table contains the main information about books (title, author, ISBN, location).
reservations	The table contains reserved books, including the issue date and return date.

DESCRIPTION OF DEPENDENCES

INTER-MODULE DEPENDENCIES

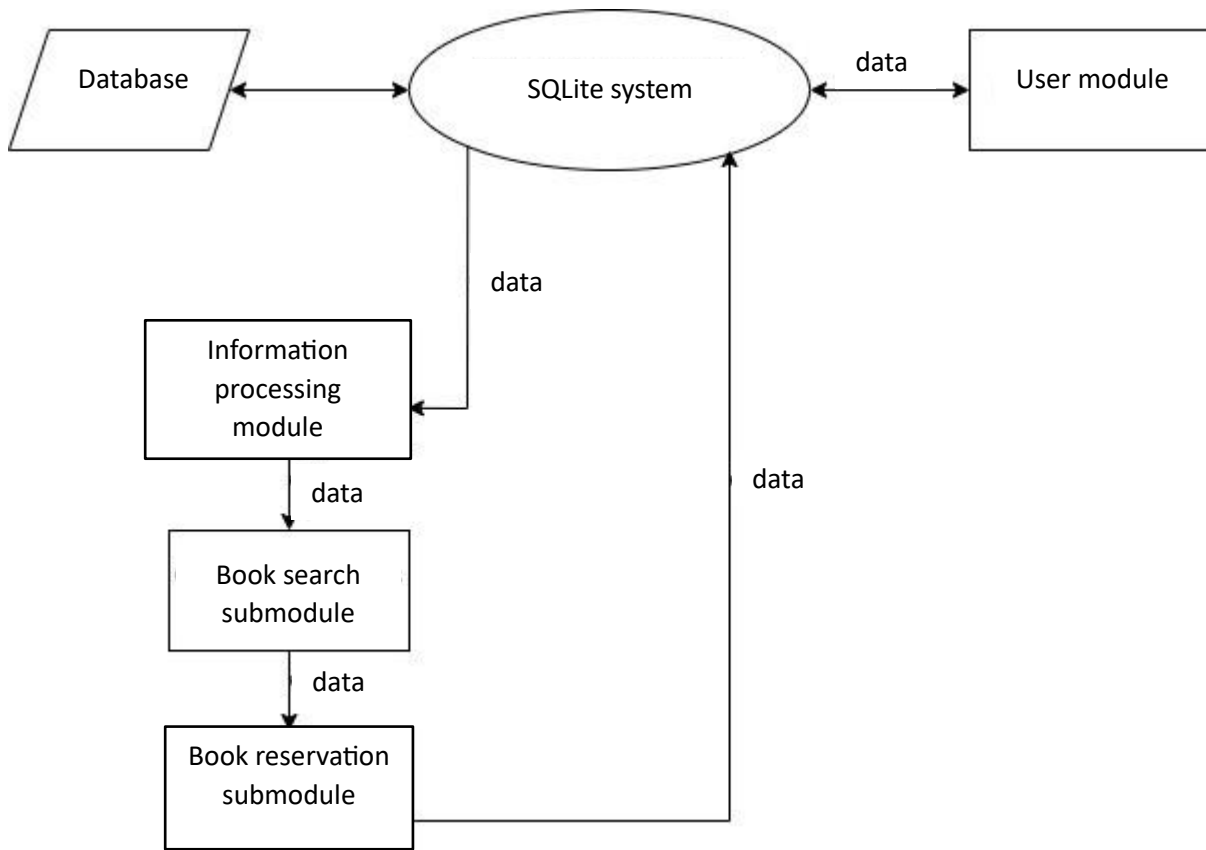


Image 2 - Inter-module dependencies

INTER-PROCESS DEPENDENCIES

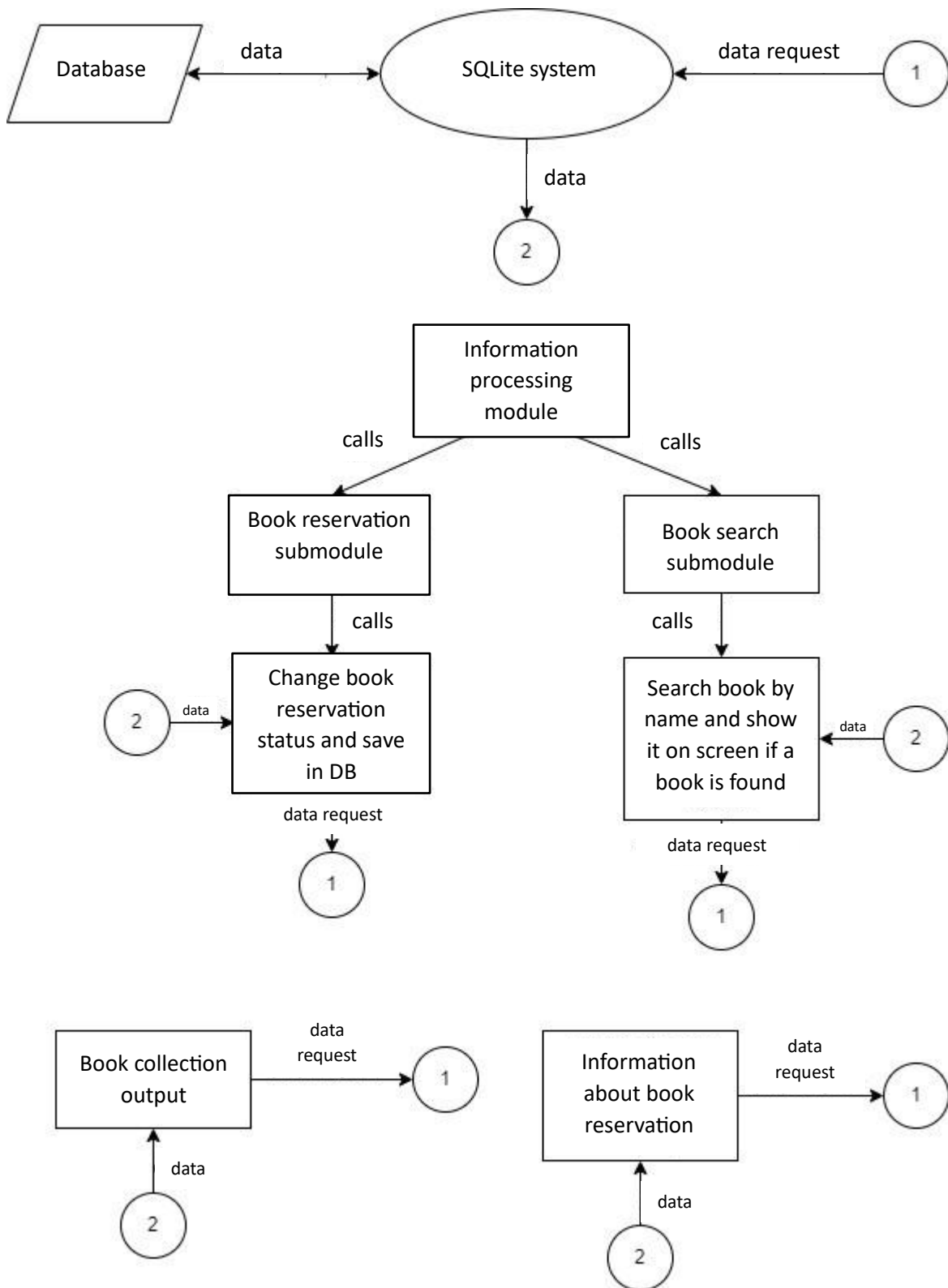


Image 3 - Inter-process dependencies

DATA DEPENDENCY DESCRIPTION

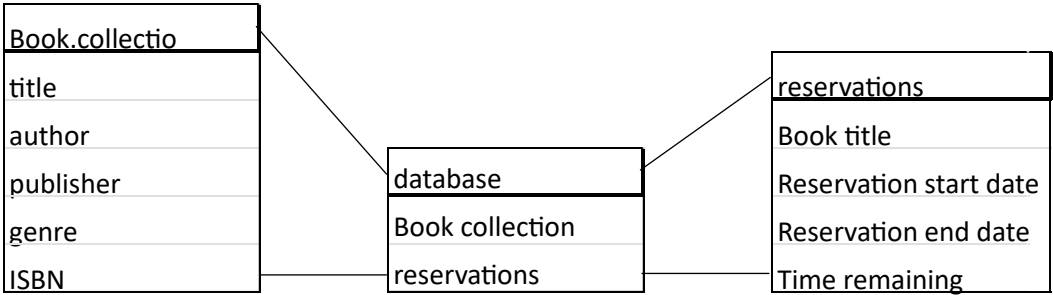


Image 4 - Data dependency description

INTERFACE DESCRIPTION

MODULE INTERFACES

Table 12

Module Name	Book Search Module
Input Data	Book title / ISBN number
Output Data	Book information

Table 13

Module Name	Book Reservation Module
Input Data	None
Output Data	Confirmation message of a successful reservation and the reservation end date/time.

PROCESS INTERFACES

Table 14

Process Name	Book Search in the Database
Input Data	*Title / ISBN
Output Data	*Book information

Table 15

Process Name	Book Reservation
Input Data	*ISBN / Title (press button)
Output Data	*Reservation confirmation message and reservation duration

USER INTERFACE

BIBLIOTĒKAS KOLEKCIJA

Q MEKLĒT

☐ PĒC NOSAUKUMA

☐ PĒC ISBN

Image 5 - Homepage initial look

BIBLIOTĒKAS KOLEKCIJA

Q ...

GRĀMATA NAV ATRASTA

Image 6 - If the book is not found

BIBLIOTĒKAS KOLEKCIJA

Q ...

ATĒLS

NOSAUKUMS:
AUTORS:
ISBN:
INFO:

STATUS: PIEEJAMA

REZERVET

Image 7 - Book found and available for reservation

BIBLIOTĒKAS KOLEKCIJA

Q...

 NOJĀUKUMS:
AUTORS:
ISBN:
INFO:

STATUS: REZERVĒTA ATLIKUSĀIS LAIKS
...

Image 8 - The book has been found and reserved

DETAILED DESIGN DESCRIPTION

DECISION TABLE

Website check

Table 16

	Conditions	1	2
C1	Is the website working?	0	1
	Actions		
D1	Display message: "Website is not working"	X	
D2	Grant access to the system		X

Checking search information

Table 17

	Conditions	1	2	3	4
C1	Was the book title entered?	0	–	1	–
C2	Was the ISBN entered?	–	0	–	1
	Actions				
D1	Display message: "No information entered for search"	X	X		
D2	Start search			X	X

Checking the search result

Table 18

	Conditions	1	2
C1	Was a record found in the database?	0	1
	Actions		
D1	Display message: "The book is not in the library collection"	X	
D2	Display book information		X

Reservation permit

Table 19

	Conditions	1	2
C1	Is the book reserved?	0	1
	Actions		
D1	The "Reserve" button appears	X	
D2	The button does not appear / remaining reservation time is displayed		X